

Solutions- Quiz 1 PCs:

$$1) y(t) = \int_{t-10}^t \cos(\tau) x(\tau) d\tau.$$

$$\text{Linearity: } y_1(t) + y_2(t) = \alpha \int_{t-10}^t \cos \tau x_1(\tau) d\tau + \beta \int_{t-10}^t \cos \tau x_2(\tau) d\tau \\ = \int_{t-10}^t \cos \tau (\alpha x_1(\tau) + \beta x_2(\tau)) d\tau \rightarrow \text{Linear}$$

$$\text{Time Invariance: } y'(t-t_0) = \int_{t-t_0-10}^{t-t_0} \cos \tau x(\tau-t_0) d\tau$$

$$y(t-t_0) = \int_{t-t_0-10}^{t-t_0} \cos \tau x(\tau) d\tau.$$

$y'(t-t_0) \neq y(t-t_0) \Rightarrow$ Not Time Invariant, not LTI.

$$2. \langle u_p, v_p \rangle = \int_{-\infty}^{\infty} u_p(t) v_p(t) dt$$

$$(a) \int_{-\infty}^{\infty} (u_c(t) \cos 2\pi f_c t - u_s(t) \sin 2\pi f_c t) (v_c(t) \cos 2\pi f_c t - v_s(t) \sin 2\pi f_c t) dt$$

$$= \int_{-\infty}^{\infty} u_c(t) v_c(t) \cos^2 2\pi f_c t dt + \int_{-\infty}^{\infty} u_s(t) v_s(t) \sin^2 2\pi f_c t dt \quad (\text{other term} \\ = 0, \text{ need to explain why})$$

$$= \int_{-\infty}^{\infty} \frac{u_c(t) v_c(t)}{2} dt + \int_{-\infty}^{\infty} \frac{u_s(t) v_s(t)}{2} dt \quad (\text{other terms zero, need} \\ \text{to explain why})$$

$$= \frac{1}{2} [\langle u_c, v_c \rangle + \langle u_s, v_s \rangle]$$

$$\left(\int_{-\infty}^{\infty} x(t) \cos 4\pi f_c t dt = 0 = x(0) \right) \\ \downarrow \text{passband}$$

b. $s_{p1}(t), s_{p2}(t)$. First find $s_1(t)$ & $s_2(t)$.? $s_1(t) = \text{Re} \{ \sqrt{\frac{2}{T}} (u(t) - u(t-T)) e^{j2\pi f_c t} \}$
 (lly for $s_2(t)$)

Turns out $\langle s_1, s_2 \rangle \neq 0$, but $\langle s_{p1}, s_{p2} \rangle = 0$.

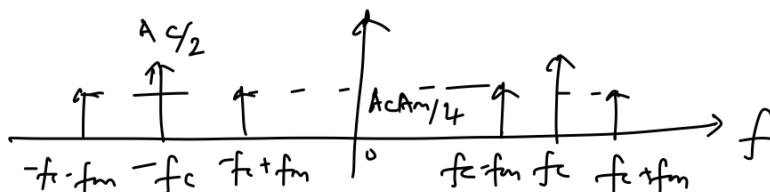
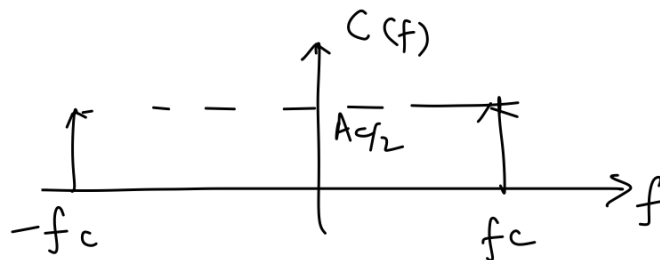
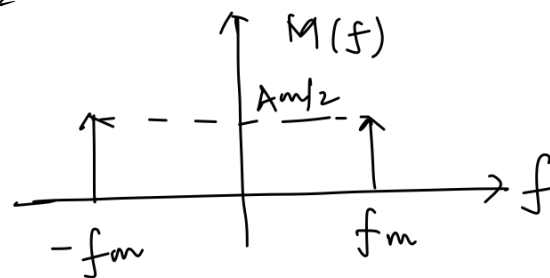
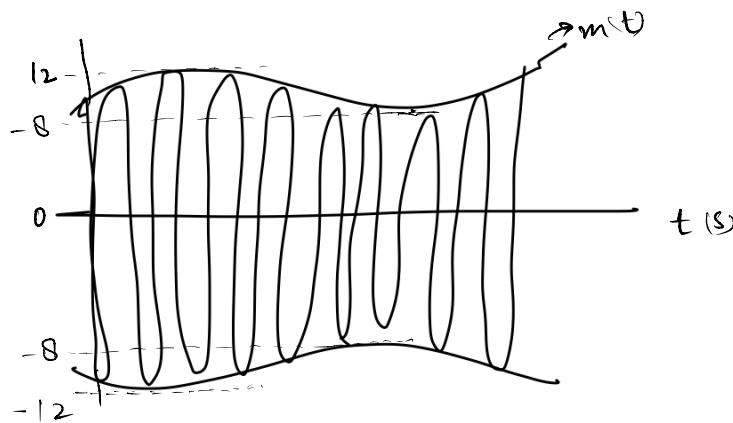
Orthogonality in baseband $\Rightarrow$ orthogonality in passband.

" " passband $\nRightarrow$ " " baseband

$$3. \quad S(t) = [A_c + A_m \cos(2\pi f_m t)] \cos 2\pi f_c t$$

$$\begin{aligned} (a) \quad S(t) &= A_c \cos 2\pi f_c t + A_m \cos 2\pi f_m t \cos 2\pi f_c t \\ &= A_c \cos 2\pi f_c t + \underbrace{\frac{A_m}{2} \left[\underbrace{\cos 2\pi (f_c - f_m) t}_{\text{LSB}} + \underbrace{\cos 2\pi (f_c + f_m) t}_{\text{USB}} \right]}_{\text{USB}} \end{aligned}$$

(b)



$$\begin{aligned} 1) \quad d) \quad \mu &= \frac{A_m}{A_c} \\ &= \frac{2}{10} = 0.2 \end{aligned}$$

$$4. \quad v_p(t) = I[-2, 2](t) \cos 100\pi t$$

$$h_p(t) = I[0, 2](t) \sin 100\pi t.$$

$$\text{In complex BB: } v(t) = I[-2, 2](t)$$

$$h(t) = -j I[0, 2](t)$$

Convolve: $v(t) * h(t)$: Trapezoid, linearly increasing between -2 to 0 , constant between 0 & 2 & linearly decreasing between 2 & 3 .

$$\text{Let } S(t) = v(t) * h(t) \Rightarrow y(t) = S(t)/2.$$

$$\text{Filter output} = \operatorname{Re}\left(\frac{S(t)}{2} e^{j2\pi f_c t}\right) = S(t) \sin 100\pi t = y_p(t)$$

(b) $y(t) \cos 100\pi t$ is passed through LPF.

Any signal at same ref freq (here 50Hz) passed through LPF gives: $v_p(t) \cos 100\pi t \rightarrow v_p(t)/2$.

$$\text{Here } y_p(t) = S(t) \sin 100\pi t$$

$$5. \quad m(t) = 3 \cos 2\pi t + 4 \sin 6\pi t \quad f_c = 600 \text{ kHz}$$

$$1. \quad \mu \Rightarrow 50\% \quad \text{or} \quad \mu = 0.5.$$

$$V_{AM}(t) = A(1 + \mu m_n(t)) \cos 2\pi f_c t$$

Need to find $m_n(t)$.

$$m_n(t) = \frac{m(t)}{M_0}, \text{ where } M_0 \text{ is the min value of } m(t).$$

$$\text{Consider } g(x) = 3 \cos x + 4 \sin 3x.$$

$$\frac{dg(x)}{dx} = -3 \sin x + 12 \cos 3x = 0.$$

$$\Rightarrow 4 \cos 3x = \sin x \quad (\text{min or max condition})$$

$$\text{Substituting above: } g(x) = 3 \cos x + 4 \cos 12x. \quad (\text{min or max})$$

$$\text{By inspection, max occurs at } x=0, \quad g(0) = 7$$

$$\text{min occurs at } \pi/12: \quad 3 \cos \pi/12 + 4 \cos \pi$$

$$= -4.25.$$

$$\Rightarrow M_0 = 4.25. \quad \& \quad m_n(t) = \frac{3}{4.25} \cos 2\pi t + \frac{4}{4.25} \sin 6\pi t$$

$$m_n(t) = 0.706 \cos 2\pi t + 0.941 \sin 6\pi t$$

$$2. \quad \text{Power efficiency: } \eta_{AM} = \frac{\tilde{m}_n^2(t)}{1 + \tilde{m}_n^2(t)}$$

$$\text{Power in } m_n(t) = \frac{(0.706)^2}{2} + \frac{(0.941)^2}{2} \approx 0.5 + 0.885 = 1.385$$

$$\eta_{AM} = \frac{1.385}{2.385} = 0.5807 \quad \text{or} \quad \text{Power efficiency is } 58.07\%.$$

3. Amplitude spectrum :

